Engineering, Construction and Architectural Management

Digital twin-enabled 3D near-misses prediction method based on trajectories and postures for proactive construction safety management

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Received 13 May 2025
Revised 28 July 2025
6 October 2025
Accepted 23 November 2025

Abstract

Purpose – This paper aims to develop a proactive construction safety management framework capable of predicting near-miss incidents by leveraging Digital Twin (DT) technology. The Digital Twin (DT) framework focuses on integrating environmental and human-related risk factors – particularly the trajectories and behaviors of workers – before accidents occur.

Design/methodology/approach – The paper proposes a computer vision-based digital twin framework that integrates real-time data on workers' movements, postures, and interactions to construct a dynamic 3D model of the construction site. Deep learning algorithms are employed to predict human trajectories and behaviors, while collision detection techniques are used to identify potential near-miss scenarios. Field validation was conducted in three high-risk zones: forklift paths, floor openings, and material lifting areas.

Findings – The proposed framework achieved high accuracy in predicting near-miss events and provided timely early warnings, outperforming traditional 2D monitoring methods. The system's capability to visualize worker interactions and spatial hazards in 3D significantly enhanced situational awareness and safety decision-making on site.

Practical implications – By transitioning from 2D to 3D monitoring and from post-event detection to real-time prediction, the proposed system enables early identification of near-miss risks during ongoing construction activities. Its integration of 3D visualization and digital twin technology provides intuitive, actionable insights, empowering safety managers to intervene proactively and enhance on-site safety performance.

Originality/value – This paper introduces an innovative approach that transitions safety management from passive detection to proactive prediction. By integrating digital twin technology with deep learning, it enables 3D visualization and prediction of near-miss incidents and supports real-time, data-driven safety interventions. The proposed system contributes to the advancement of intelligent, predictive safety management solutions in the construction industry, with a particular focus on forecasting near-miss events before they escalate into actual accidents.

Keywords Proactive safety management, Near-misses prediction, Digital twin, Computer vision

Paper type Research article

1. Introduction

The construction industry is one of the most hazardous sectors globally, characterized by high risks of accidents, injuries, and fatalities. Statistical data underscores the severity and persistence of the issue. For instance, China's Ministry of Emergency Management reported

Funding: This research is supported by Key Technologies R&D Program (Grant No. 2024YFC3808602), supported by Shenzhen Science and Technology Programs (Grant No. KJZD20230923114310021, Grant No. JCYJ20241202123722029).

Engineering, Construction and Architectural Management
© Emerald Publishing Limited
e-ISSN: 1365-232X
p-ISSN: 0969-9988
DOI 10.1108/ECAM-05-2025-0732

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1,752 fatal construction accidents in 2018 (Ministry of Emergency Management of the People's Republic of China, 2018), while in the first quarter of 2024, the Ministry recorded 3,570 production safety accidents and 3,359 related deaths across industries, marking year-on-year declines of 36.9% and 27.9%, respectively (Huaxia, 2024). Similarly, the US Occupational Safety and Health Administration recorded 1,008 construction-related fatalities in 2018, accounting for 20.4% of all occupational fatalities nationwide (Bureau of Labor Statistics, 2019). More recently, the US Bureau of Labor Statistics reported 5,283 fatal work injuries in 2023 – a 3.7% decrease from 5,486 in 2022 – indicating ongoing safety challenges despite incremental improvements (Bureau of Labor Statistics, 2023). These figures highlight the continued urgency for effective and proactive safety management systems to mitigate risks and protect workers in construction environments.

Traditionally, construction safety management has relied on reactive measures, focusing on incident response rather than prevention (Zhang *et al.*, 2025a). However, this approach has proven insufficient in addressing the root causes of accidents (Abdul Nabi and El-adaway, 2022). In recent years, the industry has shifted toward proactive strategies, emphasizing the anticipation and mitigation of risks before incidents occur (Yang *et al.*, 2012; Zeng *et al.*, 2023). Central to this proactive approach is the ability to predict accidents with high precision (Kunreuther *et al.*, 2004), which not only helps in understanding accident trends but also in evaluating the effectiveness of safety management measures. Near-misses, defined as events that could have resulted in accidents but did not (Abdelhamid and Everett, 2000), serve as early warning signs of potential hazards. These incidents have been widely utilized in various industries, such as aviation, manufacturing, and healthcare, to enhance safety management. Near-miss incidents occur far more frequently than actual accidents, often by one or two orders of magnitude (Van Der Schaaf, 1995; Zhou and Irizarry, 2016). This abundance of data makes near-misses a valuable resource for statistical monitoring and trend analysis (Ancione and Milazzo, 2025), offering opportunities to identify patterns and predict potential hazards. By leveraging near-miss data, safety management can transition from a reactive to a preventive framework, significantly reducing the likelihood of catastrophic outcomes.

However, current near-miss management practices are limited in their scope. Most approaches to near-miss management primarily focus on monitoring (Amaechi *et al.*, 2024; Salem and Dragomir, 2023; Ullah *et al.*, 2024a, b; Yan *et al.*, 2025), enabling alerts when near-misses occur, but they lack the capability to predict near-misses before they happen. Existing predictive approaches, including scenario analysis, regression models, grey systems, time-series analysis, Markov chains, Bayesian networks, and neural networks, offer varying degrees of effectiveness for near-miss forecasting in construction (Das Chagas Moura *et al.*, 2016; Doss *et al.*, 2020; Li *et al.*, 2016; Pljakić *et al.*, 2019; Radaideh *et al.*, 2020; Tang *et al.*, 2016; Zheng and Liu, 2009). While these methods can provide predictions, they are largely based on historical data, using past patterns to infer future probabilities rather than dynamically simulating and predicting imminent 3D interactions between workers, machinery, and the environment as they evolve in real time. Consequently, they fail to provide the ability to actively defend against potential near-misses during the construction process. To address this gap and enable proactive safety measures, it is critical to predict the three-dimensional interactions between workers and the construction environment in real time.

Digital twin technology has emerged as a powerful tool for enabling 3D near-miss prediction and proactive safety management. A digital twin is a virtual replica of a physical construction environment, capable of real-time monitoring, simulation, and predictive analysis (Liu *et al.*, 2021, 2022; Ye *et al.*, 2023; Zhao *et al.*, 2022). By integrating data from sensors, computer vision, and other sources, digital twins provide spatially accurate, dynamic representations of construction sites. This capability allows for the visualization of potential near-miss incidents in three dimensions, offering a comprehensive understanding of spatial relationships and hazards.

In this context, this study proposes a proactive safety management system that leverages digital twin technology and an AI prediction engine to anticipate near-miss incidents on

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construction sites. Unlike traditional approaches such as GPS or BIM, which are constrained by static modeling or dependence on wearable sensors, the DT-based framework offers a dynamic, non-intrusive, and predictive solution tailored to the complexity of construction environments. The system utilizes computer vision to construct a digital reconstruction of the construction environment, enabling real-time monitoring of worker activities. By detecting keypoints of construction personnel, the system predicts workers' trajectories and postures, facilitating highly detailed 3D near-miss predictions. The immersive 3D visualization environment, developed using Unreal Engine, enhances situational awareness and supports intuitive decision-making.

The system's effectiveness was validated through field tests conducted in three high-risk areas: the forklift path zone, the floor opening zone, and the material lifting zone. While these scenarios represent typical and critical risk environments, construction sites are inherently complex and diverse. Therefore, future work will involve more extensive validation across different types and scales of construction projects to further ensure the system's applicability and effectiveness. Such efforts aim to enhance the generalizability of the proposed framework and provide a more universal solution for safety management in the construction industry.

This approach not only enhances the accuracy of near-miss detection but also supports proactive safety management by preventing incidents before they occur. The proposed system represents a significant advancement in construction safety, offering a novel framework for reducing risks and improving overall site safety.

2. Related works

2.1 Near-miss management: from traditional to technological approaches and the remaining challenges

The critical role of near-miss data in proactive construction safety management has been widely recognized, particularly given the persistently high-risk nature of the construction industry. Grounded in Heinrich's accident pyramid theory (Heinrich, 1941), which was subsequently expanded by Bird, near-miss incidents are considered fundamental to occupational safety, serving as crucial early indicators of potential hazards. Bird's extensive analysis of over 1.7 million accident reports revealed a striking ratio: for every fatality, there are approximately 600 near-miss incidents, predominantly triggered by unsafe acts (Bird and Germain, 1966). These near-miss events, often regarded as "unrealized accidents," provide valuable opportunities for intervention. Proactive management of these incidents can substantially mitigate the likelihood of severe accidents, thereby preventing more serious outcomes (Haas *et al.*, 2020).

Despite their significance, traditional approaches to near-miss data collection and analysis are often labor-intensive and heavily reliant on manual processes, which inherently limit their efficiency and scalability. This limitation has spurred the development of IT-based solutions designed to enhance near-miss detection and management. Recent advancements include the introduction of automated systems and sensor-based tools for real-time data collection and incident tracking. For example, Aria *et al.* (2014) successfully tested wearable sensor-based technologies for detecting near-miss falls, while Yang *et al.* (2014, 2015, 2016) demonstrated the potential of automated systems utilizing inertial measurement technology to identify hazardous movements. Zhang *et al.* (2019) further developed innovative models integrating mobile technologies and artificial neural networks to identify fall-related near misses. Teizer and Cheng (2015) proposed proximity detection models that employ automated location tracking to assess worker proximity to equipment and site layout hazards. More recently, Wu *et al.* (2010) introduced real-time RFID-based tracking systems, and Chung *et al.* (2023) explored IoT-based monitoring solutions for real-time near-miss detection. Nasernejad *et al.* (2021)

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advanced this field by using video data and a Gaussian Process IRL approach to model pedestrian behavior in vehicle near-miss scenarios.

Significant progress has also been made in the prediction of near-miss incidents. For instance, Kataoka *et al.* (2020) developed a pedestrian near-miss detector with temporal analysis to provide both pedestrian detection and risk-level predictions, showcasing advancements in pedestrian safety technology. Ding *et al.* (2020) introduced a Conditional Multiple Trajectory Synthesizer framework to generate near-miss data using Variational Bayesian methods, enhancing trajectory prediction accuracy for risky driving scenarios. Workineh *et al.* (2023) focused on developing a maternal near-miss prediction model among pregnant women, aiming to generate a risk score for maternal near-miss incidents. Additionally, Madsen *et al.* (2024) utilized near-miss data, artificial intelligence, and machine learning to predict maritime incidents, demonstrating the utility of recent near-miss counts in predicting future serious casualties at the waterway level. However, these predictive efforts primarily focus on analyzing and forecasting near-miss data without addressing the dynamic processes leading to near-miss events, failing to address the dynamic, three-dimensional interactions that drive most construction site hazards. This limitation hinders the ability to anticipate near-misses during their occurrence, thereby preventing truly proactive safety management.

In summary, while significant advancements have been made in near-miss detection, data collection, and prediction, critical challenges remain in the construction sector. Traditional methods are limited by their reliance on manual processes, and although IT-driven solutions have improved efficiency, most current approaches are still two-dimensional (operating in planar (x, y) coordinates, lacking vertical (*z*-axis) analysis, thus missing hazards like overhead crane interactions or falls from height), failing to fully capture the complex, three-dimensional dynamics of construction sites. The chaotic and dynamic nature of construction environments further complicates near-miss prediction, as constant site changes and intricate worker-equipment interactions are difficult to model accurately. Moreover, existing predictive models, while innovative, often focus on post-hoc analysis rather than real-time process prediction. To address these limitations, there is a pressing need for the development of three-dimensional (integrating x, *y*, *z* axes to model 3D positions, trajectories, and postures, capturing multi-directional interactions like workers near floor openings or under suspended loads), granular near-miss prediction methods that can monitor and forecast incidents in real-time. Such advancements would enable truly proactive safety management, ultimately reducing the risk of severe accidents and enhancing overall safety in the construction industry.

2.2 The capacity of digital twin in proactive safety management

Digital twin (DT) technology, which comprises a physical entity, a digital counterpart, and an integrated data connection, has emerged as a transformative tool for enhancing the safety and efficiency of complex physical systems (Castilla *et al.*, 2024; Feng *et al.*, 2025). Initially conceptualized during NASA's Apollo program and later formalized by Michael Grieves, DT technology replicates the lifecycle of a physical system in a digital environment, enabling real-time simulation and predictive analytics (Grieves and Vickers, 2017). By continuously synchronizing with sensor and environmental data, digital twins can dynamically model the current state of physical assets, thereby identifying potential failures before they occur.

The application of DT technology has expanded significantly beyond equipment monitoring, encompassing sectors such as manufacturing, logistics, energy, and construction (Martinez *et al.*, 2025; Xia *et al.*, 2024). For instance, Rauscher *et al.* (2021) demonstrated its utility in nuclear reactor maintenance by developing a simulation tool that reduces downtime and enhances decision-making. In logistics and robotics, Wu *et al.* (2023) and Hu (2022) employed DT frameworks, including the Internet of Everything (IoE) and mutual information algorithms, to optimize operations and ensure the continuity of complex systems. In the construction sector, DT has been primarily utilized to create accurate 3D digital

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replicas of infrastructure, facilitating progress tracking, asset management, and digital fabrication. For example, Long *et al.* (2024) emphasized how high-fidelity DT models can improve project management and the oversight of complex construction assets. Similarly, Kang *et al.* (2021) applied DT for bridge health monitoring, leveraging sensor networks to synchronize real-time data between physical and virtual models, thereby enhancing maintenance efficiency and extending asset longevity.

Recently, DT technology has gained traction for predictive applications across various fields. Park *et al.* (2021) developed a DT model to predict the steady-state temperature field of a natural gas boiler, while Jiang *et al.* (2021) proposed a DT-driven framework for fatigue life prediction in steel bridges using a probabilistic multiscale model. He *et al.* (2021) reviewed DT-driven methods for predicting the remaining useful life (RUL) of gear performance degradation, focusing on physical and virtual model-based approaches. In the transportation sector, Hu *et al.* (2022) introduced a DT-assisted real-time traffic data prediction method for 5G-enabled Internet of Vehicles, demonstrating its feasibility in smart traffic flow and velocity prediction. Katsidoniotaki *et al.* (2022) developed a DT for predicting extreme loads on a wave energy conversion system, showcasing its innovative potential in renewable energy. Additionally, Wang *et al.* (2023) established a DT model for gas pipeline leakage detection based on pressure signals, and Liu *et al.* (2023) presented an intelligent DT prediction and reverse control system architecture for thermal errors, enabled by deep learning and cloud-edge computing. In the construction domain, Jiao *et al.* (2024) proposed a DT framework utilizing cloud-edge cooperation to predict worker trajectories, balancing computational efficiency with data privacy concerns.

Despite these advancements, the application of DT technology in construction sites faces significant challenges (Lu and Antwi-Afari, 2024; Luo *et al.*, 2024). The chaotic and dynamic nature of construction environments, characterized by constantly changing workers, materials, and machinery, complicates the creation of accurate virtual representations. Furthermore, the predictive capabilities of DT systems in proactive safety management remain in their infancy, with most systems still limited to monitoring functions (Chung *et al.*, 2023; Teizer and Cheng, 2015; Wu *et al.*, 2010; Yan *et al.*, 2025). These systems primarily focus on real-time data collection and incident tracking rather than forecasting potential risks before they occur, failing to enable proactive safety interventions in dynamic construction environments. To fully realize the potential of DT in proactive safety intervention, challenges related to dynamic modeling, sensor data integration, and the incorporation of machine learning algorithms must be addressed. Overcoming these barriers will enable construction teams not only to monitor real-time conditions but also to predict potential risks and hazardous events, thereby transforming safety management from a reactive to a truly proactive approach.

3. Methodology

3.1 Framework of digital twin system

To enable real-time prediction of near-miss incidents during construction processes, it is essential to model and evaluate the interactions between construction workers and their surrounding environment (e.g. machinery, materials, and structures) for safety risk identification. This study leverages digital twin technology to dynamically reconstruct construction sites in a virtual environment, facilitating real-time simulation of interactions among on-site elements. For construction workers specifically, their motion trajectories and behavioral patterns are predicted in the digital twin domain by analyzing human keypoints extracted from pose estimation algorithms. This approach enables proactive detection of near-miss events at their early stages. Moreover, the digital twin platform visualizes predicted near-miss scenarios, including spatial relationships and temporal sequences, providing construction managers with actionable insights. Such visualization enhances data-driven decision-making, supports timely interventions, and strengthens proactive safety management in construction.

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Based on the concept of digital twins, the digital twin system comprises three components: the physical layer, digital layer, and connectivity layer. The framework of the digital twin system employed in this study is illustrated in the Figure 1.

To accurately reconstruct construction site elements within the digital twin environment, different reconstruction strategies are applied based on the nature of the elements. Elements that remain relatively unchanged over short periods, such as the main construction structure and stored materials, are categorized as static elements and updated periodically. In contrast, elements that undergo continuous changes, such as moving workers, construction machinery, and actively used materials, are classified as dynamic elements and updated in real time.

The workflow of the digital twin system (as shown in Figure 2) is as follows: Initially, static data from the construction site is collected using drones, while real-time dynamic information – including dynamic construction materials, machinery, and personnel – is captured through binocular cameras deployed on-site. These data are then transmitted to the data processing module for analysis. Through algorithms for object recognition, object tracking, binocular stereo matching, and human pose estimation, the system extracts the positional information of both static and dynamic elements, as well as the keypoint information of construction personnel.

Subsequently, this information is fed into the digital twin platform to create a digital twin model of the construction site within a virtual environment. This model not only reflects the real-time status of each construction worker but also predicts their movement trajectories and postures based on historical data using transformer-based algorithms. The system can forecast near-miss incidents and issue early warnings within the digital twin domain. Moreover, the visualization capabilities of the digital twin platform provide a more intuitive basis and assurance for safety management on the construction site.

The digital twin (DT) system developed in this study is applicable to construction scenarios that feature high spatial complexity, dynamic environmental changes, and stringent safety requirements. Its applicability is guided by the following three principles:

(1) Presence of complex three-dimensional environments

The system is particularly suited to construction sites with multi-level structures, scaffoldings, and overlapping operational zones, where traditional 2D safety systems cannot capture vertical

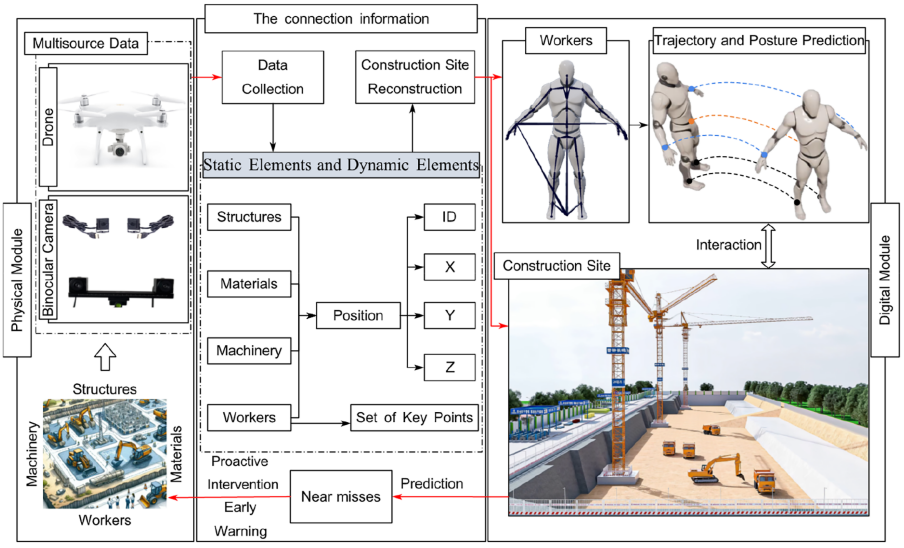

Figure 1. Framework of the digital twin system. Source: Authors' own work

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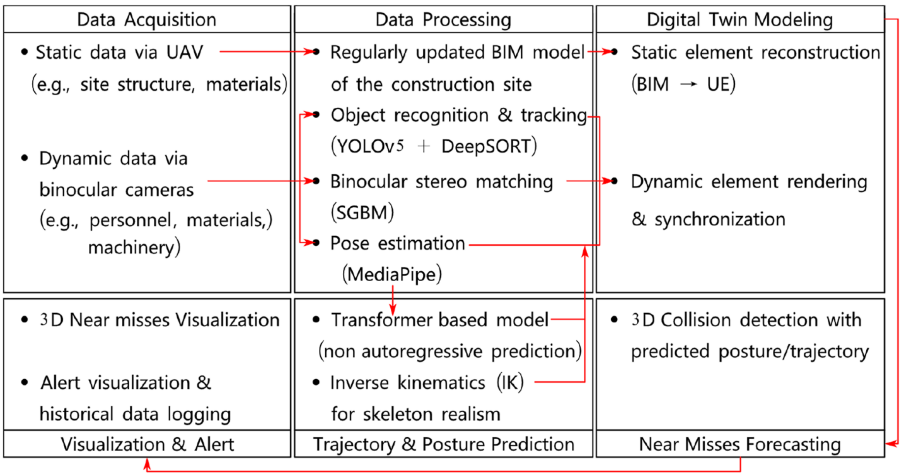

Figure 2. Workflow of the proposed digital twin method. Source: Authors' own work

risks or spatial layering. The DT framework enables full 3D modeling and analysis, making it ideal for environments where elevation and depth relationships are safety-critical.

(2) Frequent dynamic changes

Construction environments are inherently dynamic, with frequent changes in equipment layout, material locations, and worker movement patterns. The proposed DT system is intended for use in scenarios where static monitoring tools (e.g. BIM snapshots or fixed sensor thresholds) are insufficient to reflect real-time site conditions.

(3) Tasks involving high safety risks and requiring proactive intervention

The system is especially beneficial in high-risk operational zones such as crane paths, material hoisting areas, or floor openings, where passive alerts may be too late. Its predictive capability – based on trajectory and posture forecasting – supports early prediction of potential near-miss events and enables preemptive safety actions.

3.2 Physical layer description

The physical layer forms the foundational data acquisition component of the digital twin system, enabling precise replication of the construction environment. To address the dynamic and multifaceted nature of construction sites, this system integrates unmanned aerial vehicles (UAVs) and binocular cameras for robust and comprehensive data collection. UAVs capture static features, such as structural elements and material storage zones, while binocular cameras provide continuous monitoring of dynamic entities, including workers, machinery, and moving materials.

The successful implementation of the physical layer relies on environmental and technical conditions. The system performs optimally under stable weather, adequate lighting, and carefully planned sensor deployment. For example, UAV operations require clear skies and minimal wind to ensure stable flight and high-resolution imagery. Similarly, binocular cameras achieve the best results when placed at elevated positions with wide, unobstructed views, and when deployed during daylight hours with consistent lighting. The subsequent parts elaborate on the deployment strategies and operational considerations for these sensing technologies.

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UAVs play a critical role in periodically mapping static elements of the construction site, ensuring an accurate and current depiction of project progress. Several key factors guide their deployment and data acquisition:

(1) Data Collection Frequency: UAV scanning intervals should synchronize with the construction timeline. A weekly scan is typically sufficient for tracking structural updates, though additional flights may be warranted during pivotal phases, such as concrete pouring or major installations.

(2) Flight Altitude and Path Optimization: Flight altitudes, generally set between 50 and 100 meters, must strike a balance between coverage breadth and image resolution. Automated path planning is essential to achieve comprehensive site coverage and minimize data gaps from inadequate overlap.

(3) Lighting and Weather Constraints: To ensure consistent image quality and avoid issues from shadows or fluctuating light, UAV operations should occur during stable lighting periods (e.g. 9–11 AM or 2–4 PM). Flights should also be halted during adverse weather – such as high winds, rain, or snow – to maintain stability and safeguard data integrity.

Binocular cameras facilitate real-time observation of dynamic construction activities by tracking the movements of personnel, equipment, and materials. Effective deployment hinges on considerations of visibility, occlusion risks, and light conditions:

(1) Effective camera placement is critical for optimal site coverage. Mounting binocular cameras on tower cranes or elevated structures provides a wide, unobstructed vantage point across the construction site. Strategic positioning in high-activity zones – such as areas for formwork installation, steel reinforcement, or concrete pouring – enables targeted observation of critical operations. Cameras should also be installed near heavy machinery, including excavators, cranes, and concrete mixers, to monitor worker-equipment interactions, with placement adjusted to avoid disrupting workflow.

(2) Deployment Environment: To optimize performance, binocular cameras should be deployed in environments with favorable lighting conditions. Operations are effective during periods of consistent, ample illumination – such as mid-morning (e.g. 9–11 AM) or early afternoon (e.g. 2–4 PM) – to ensure high-quality image capture without reliance on supplemental enhancements. Deployment should be planned to avoid areas or times with poor natural light, minimizing the risk of degraded visibility or data inconsistency.

By combining regularly updated drone-collected static data with real-time dynamic data from binocular cameras, the physical module creates a comprehensive and up-to-date representation of the construction site. This robust data foundation is transmitted to the data processing module, where it undergoes further analysis and integration into the digital twin platform.

In addition to enabling accurate environmental reconstruction, the integration of UAVs and binocular cameras within the digital twin framework also contributes to improved safety planning and equipment deployment strategies on site. Specifically, the virtual construction environment modeled by the digital twin allows for pre-deployment simulation of UAV flight paths and camera placements. This enables early identification of potential risks such as collision with cranes or scaffolding, occlusion by temporary structures, or insufficient visual coverage of high-risk zones. Furthermore, digital twin simulation supports the optimization of UAV operation windows – accounting for lighting conditions, wind speed, and site congestion – to reduce the likelihood of flight instability or data collection failures. For binocular cameras, the system can be used to test various mounting configurations to avoid obstructing

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construction workflows and ensure continuous monitoring of safety-critical activities. These lessons reinforce the value of digital twin technology not only in enhancing sensing accuracy but also in improving the safe and effective deployment of site monitoring equipment.

3.3 Digital twin implementation

This section specifically details the implementation plan of the digital twin system for the construction site. It covers the mapping plan from the physical world to the digital twin virtual world to conduct a complete digital reconstruction of the construction site. Additionally, it presents the prediction scheme for the trajectories and postures of reconstructed construction personnel, the determination method for near-misses on the construction site, and the realization of near-misses event prediction. The entire digital twin system was developed using Unreal Engine 5 (UE5), a powerful real-time 3D engine known for its high-fidelity visualization capabilities and extensibility for simulation-based applications. The official documentation of UE5 is available at: https://docs.unrealengine.com/5.0/en-US/.

3.3.1 Physical state information mapping. To achieve more accurate reconstruction of multi-source information on the construction site, we adopt distinct reconstruction strategies for static and dynamic elements.

For static data collected by drones, after information extraction in the data processing module, as shown in Figure 3, the data is used to construct a static Building Information Model (BIM). This BIM model is then imported into the digital twin system developed in Unreal Engine (UE) for reconstruction and is updated regularly in accordance with the construction progress.

For dynamic data collected by binocular cameras deployed on the construction site, the data processing module extracts real-time updated IDs of dynamic construction personnel, machinery, and materials, along with their coordinates (x, y, z) in the world coordinate system with the camera as the origin. Initially, the YOLOv5 algorithm (Jiang *et al.*, 2022) processes the left image to identify and classify element types, distinguishing construction workers, materials, and machinery. Concurrently, the left and right images undergo Semi-Global Block Matching (SGBM) (Hirschmuller, 2008) to derive depth information, facilitating accurate three-dimensional spatial reconstruction. Subsequently, the DeepSORT algorithm (Wojke *et al.*, 2017) is utilized to assign unique IDs to each detected object, ensuring consistent tracking across the dynamic construction environment. Additionally, for the identified

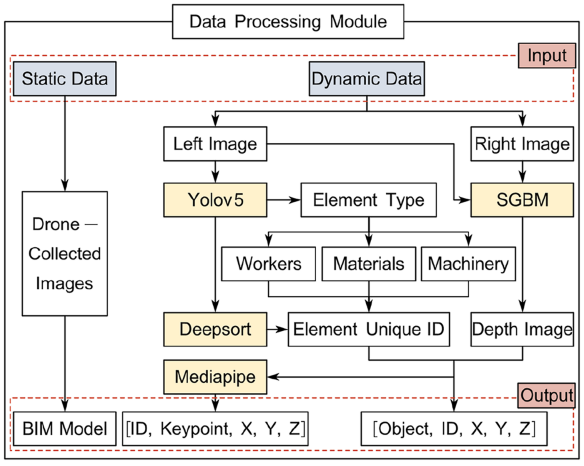

Figure 3. Workflow of the data processing module. Source: Authors' own work

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construction personnel, the module integrates MediaPipe (Lugaresi *et al.*, 2019) to extract the coordinates of key body parts, yielding precise anatomical references.

To seamlessly integrate physical entities with the digital model, the TCP/IP communication protocol is adopted, supporting full-duplex data exchange. Real-time dynamic information is transmitted to the UE-based digital twin platform for reconstruction and analysis. To streamline the real-time rendering of dynamic elements, a foundational model database – including common machinery, materials, and personnel – has been established. Upon receiving data from the physical environment, the platform queries the unique ID of each detected dynamic element. If the corresponding model exists in the digital twin, its position is updated accordingly; if absent, the system instantiates and positions the appropriate model within the virtual environment.

To date, this framework enables comprehensive reconstruction of construction site elements within the digital twin platform, accurately replicating both static and dynamic components. However, capturing the nuanced interactions between construction personnel and their surroundings with greater fidelity demands more than static models can offer. Positional tracking alone falls short; reconstructing personnel postures to reflect real-world movements is critical for a faithful digital representation. This entails driving digital avatars with motion data synchronized to onsite worker actions, a process that builds on existing tracking capabilities but introduces new complexities.

While personnel identification and three-dimensional keypoint extraction provide a foundation for posture reconstruction, directly applying these keypoints to drive a model often yields unrealistic joint rotations or joint extensions, as shown in Figure 4. Achieving effective posture alignment requires addressing several challenges inherent to mapping real human motion onto digital avatars:

(1) Inconsistency in Scale and Proportion:Inconsistency in Scale and Proportion: The scale of the human body represented by the keypoints may not align with that of the target model. Additionally, the proportions of various body segments can differ significantly.

Figure 4. Unrealistic joint rotations and joint extensions. Source: Authors' own work

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(2) Coupling Problem: There is inherent coupling among the target keypoints as well as within the human skeleton itself. This interdependence means that it is unfeasible to optimize individual parts independently; a holistic optimization approach is necessary.

(3) Joint Activity Constraints: Human joints do not possess omnidirectional movement capabilities. Each joint has specific movement constraints, some of which are soft constraints, limiting their range of motion.

To address these issues, we employ inverse kinematics (IK) (Aristidou *et al.*, 2018) to calculate the Euler rotation angles of the joints and bones of the human body based on the positions of the keypoints. The model is driven to perform human posture reconstruction by transmitting the Euler rotation angles of the bones. The IK solution is crucial for achieving more accurate posture reconstruction of construction personnel, thereby enhancing the realism and effectiveness of the digital twin system for the construction site.

Inverse Kinematics (IK) solution is a technique widely used in computer graphics and robotics to determine the joint angles of a kinematic chain given the desired end-effector position. The basic idea behind IK is to minimize an error function that measures the difference between the desired end-effector position and the actual position achieved by the kinematic chain. This is typically done using iterative optimization algorithms.

In our work, we implement an Inverse Kinematics (IK) solver based on PyTorch for solving skeletal systems. The Limited-memory Broyden-Fletcher-Goldfarb-Shanno (LBFGS) optimization algorithm (Zhang *et al.*, 2025b) is employed to iteratively minimize the error between the target keypoints of the skeletal model and the detected keypoints. The IK solver adjusts the joint angles, represented as Euler angles, within the skeletal hierarchy, ensuring accurate alignment between the bones and the target keypoints, thereby achieving precise posture reconstruction.

The skeletal system is modeled hierarchically, where each bone's local transformation matrix is defined relative to its parent bone, as shown in Figure 5. This hierarchical structure enables efficient propagation of transformations along the kinematic chain, from the root to the end-effectors (e.g. hands, feet). The local transformation matrix M_{local} describes the bone's rotation and translation relative to its parent, while the global transformation matrix M_{world} for each bone is recursively computed as:

$$M_{\text{world}} = M_{\text{parent}} \cdot M_{\text{local}} \tag{1}$$

For the root bone, which has no parent, the global transformation matrix is simply its local transformation matrix. This structure ensures that each bone's movement affects all its child bones.

The goal of IK is to adjust joint angles so that the skeletal model's keypoints match the observed keypoints as closely as possible. We use the Mean Squared Error (MSE) (Hodson, 2022) as the loss function to quantify the difference between the detected keypoints and the target keypoints:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} \left\| \mathbf{x}_i^{\text{detected}} - \mathbf{x}_i^{\text{target}} \right\|^2 \tag{2}$$

where $\mathbf{x}_i^{\text{detected}}$ is the detected keypoint position, $\mathbf{x}_i^{\text{target}}$ is the location of keypoints of the model to be optimised, and N is the total number of keypoints. Minimizing this loss allows us to optimize the joint angles to accurately reconstruct the target posture.

The optimization process is driven by the LBFGS algorithm, which is particularly suited for high-dimensional optimization problems. LBFGS approximates the inverse Hessian matrix using past gradient updates, allowing for fast convergence without the need to explicitly compute second-order derivatives. The optimization process follows these steps:

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Figure 5. The skeletal system. Source: Authors' own work

(1) Compute the gradient $\nabla\mathcal{L}$ of the loss function

(2) Update the joint angles θ using the approximate inverse Hessian:

$$\theta_{t+1} = \theta_t - \alpha H^{-1}\nabla\mathcal{L}(\theta_t) \tag{3}$$

where is α the learning rate and H is the approximate Hessian matrix.

(3) Iterate this process until the loss function converges to a predefined tolerance.

During optimization, joint constraints are enforced to ensure that the resulting poses remain physically plausible. Each joint in the human body has a specific range of motion. To enforce these limits, we use a barrier function that penalizes joint angles exceeding their allowed range:

$$\mathcal{L}_{\text{barrier}} = \sum_i \left(e^{4\left(\theta_i - \theta_{\max,i}\right)} + e^{4\left(\theta_{\min,i} - \theta_i\right)}\right) \tag{4}$$

where $\theta_{\min,i}$ and $\theta_{\max,i}$ represent the minimum and maximum permissible angles for joint i, and θ_i is the current joint angle. This term ensures that joint angles remain within valid limits by drastically increasing the loss when the joint angle exceeds the predefined bounds.

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In addition to joint constraints, we apply L_2 regularization to prevent extreme or unnatural joint rotations. The regularization term is formulated as:

$$\mathcal{L}_{\text{reg}} = \lambda \sum_i \theta_i^2 \tag{5}$$

where λ is a regularization coefficient that balances the influence of the joint angle regularization. This term helps maintain smooth and natural joint motions, ensuring that the optimization does not produce unrealistic poses.

PyTorch's automatic differentiation feature simplifies the implementation of these constraints and regularization. By constructing a computational graph during the forward pass, PyTorch automatically computes the necessary gradients and updates the joint angles efficiently using LBFGS.

To improve computational efficiency, several optimizations are introduced. First, precomputing the parent-child relationships between bones reduces redundant calculations during the forward pass. Additionally, critical matrix operations are accelerated by offloading complex calculations to a custom C++ module, significantly enhancing computational efficiency. Finally, Moving Least Squares (MLS) smoothing is applied to the joint angles and translations across consecutive frames, ensuring fluid and realistic skeletal motions. These optimizations enable the IK solver to perform fast and accurate posture reconstruction, making it suitable for real-time applications, such as monitoring and managing construction activities.

3.3.2 Prediction of the trajectory and posture. In this section, we apply the algorithm proposed by Mahdavian *et al.* (2023). This algorithm enables us to predict both the trajectory and posture of human subjects based on keypoints extracted using MediaPipe. As human motion occurs, the positions of keypoints change in a predictable, time-dependent manner – for example, the rhythmic swinging of arms and legs while walking or the stride pattern during running. These temporal dependencies capture the underlying motion patterns across successive time frames, enabling the model to understand how keypoints are likely to shift in the near future based on past trends. By leveraging the temporal dependencies within the observed keypoints, the method allows for efficient and accurate forecasts of human motion, capturing both position and limb configuration in real-time.

As shown in Figure 6 (Mahdavian *et al.*, 2023), the non-autoregressive transformer architecture aligns well with our requirements, as it predicts the entire sequence of future movements in parallel rather than step by step, thus avoiding the cumulative errors typically seen in autoregressive models. The model consists of an encoder-decoder structure with L layers, each containing multi-head attention and fully connected layers. The encoder processes 3D human poses $X^P_{1:M}$ or hip trajectories $X^T_{1:M}$, producing two embedding sequences: $Z^P_{1:M}$ and $Z^T_{1:M}$. To capture spatial dependencies, the pose and trajectory encoding networks, ϕ_P and ϕ_T, are based on Graph Convolutional Networks (GCN). These networks learn the spatial relationships between body joints, using an adjacency matrix to compute D-dimensional embeddings for the input sequences. For non-autoregressive prediction, the last frame of the input sequences X^P_M and X^T_M is copied and used as a query for the decoder. The networks, ψ_P and ψ_T, based on Multi-Layer Perceptron (MLP), generate parallel predictions for posture and trajectory, $X^P_{M+1:M+N}$ and $X^T_{M+1:M+N}$, using the decoder outputs and residual connections with query sequences derived from the last frame of the input.

A critical feature of the algorithm is the shared attention mechanism between the trajectory and posture prediction tasks. This mechanism allows the model to transfer information between these two predictions, improving overall accuracy. For example, subtle posture changes often indicate an impending change in movement direction. The shared attention mechanism identifies these cues, enabling the model to refine its trajectory predictions based on posture adjustments and vice versa.

For trajectory prediction, the model forecasts the 3D position of the human's center of mass, typically represented by the hip joint. This provides essential information on how the

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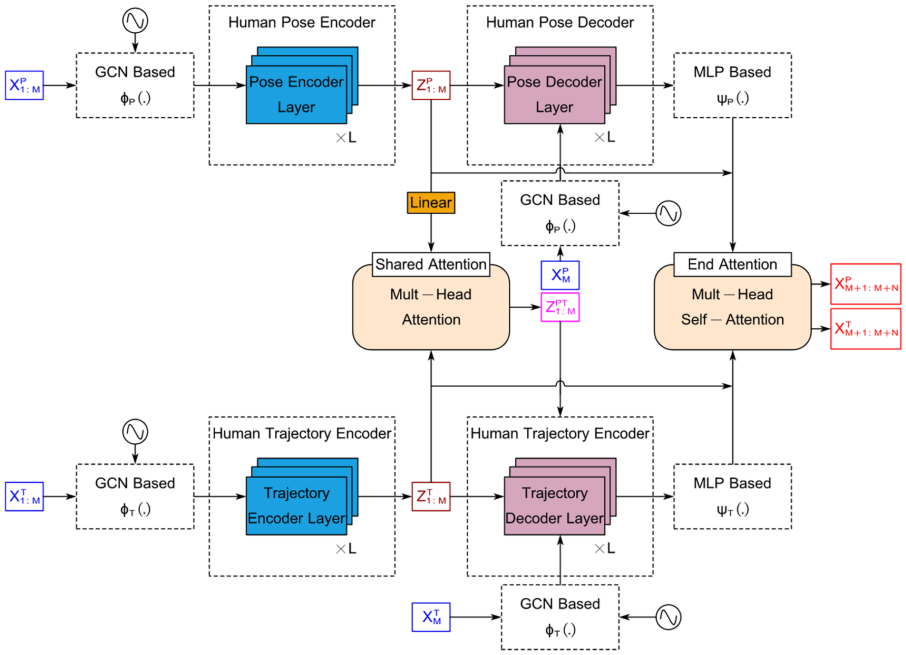

Figure 6. The structure of the algorithm. Source: Authors' own work

human body moves through space. Simultaneously, the posture prediction captures the relative positions of all other joints with respect to the hip, modeling the full configuration of the human body. By predicting trajectory and posture together, the algorithm ensures that both aspects of human movement are accurately modeled, contributing to a more realistic and comprehensive prediction of human motion.

The training process for the model utilizes a loss function that balances both trajectory and posture errors. The trajectory loss measures the difference between predicted and actual hip positions, while the posture loss evaluates the accuracy of joint angle predictions. By optimizing these two loss components simultaneously, the algorithm ensures accurate and realistic forecasts of both global (body movement through space) and local (limb configuration) dynamics. This dual-objective optimization is particularly important in construction scenarios, where both the path of worker movement and the fine-grained posture – such as bending near edges or reaching near machines – carry safety implications. In addition, this design improves model robustness in handling variations across different activities and movement speeds.

Once the detected human keypoints are fed into the trained algorithm, it returns the predicted the human trajectory (represented by the hip joint position) and keypoints in time. The predicted keypoints and trajectory are then processed through the IK solver, as described in the previous section, to compute the skeletal joint angles. These joint angles drive the character in the digital twin system established on the Unreal Engine (UE) platform, facilitating a real-time representation of human movement and posture. This workflow enables accurate prediction and visualization of construction workers' trajectories and behaviors within the digital twin, supporting safety monitoring and operational analysis on the construction site.

3.3.3 Prediction of the near-misses. By reconstructing the construction site within a digital twin framework, we create a comprehensive virtual replica of the physical environment. This

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digital twin facilitates early prediction of near-miss incidents by continuously monitoring interactions between workers and their surroundings in real time. For each dynamic object on the site – excluding construction personnel – we define a protective "envelope" that acts as a safety buffer. These envelopes, tailored to the object's characteristics, are modeled as geometric shapes such as cubes, spheres, or cylinders, as depicted in Figure 7. Similarly, static elements within the digital twin, such as structural openings, foundation pits, and unguarded edges, are assigned protective boundaries to delineate predefined hazardous zones.

The near-miss prediction method operates by assessing spatial overlaps between a worker's predicted position and any protective envelope in the digital twin. When a worker's forecasted trajectory intersects an envelope, the system flags a potential near-miss event. This predictive process unfolds dynamically as workers and objects move, allowing the digital twin to anticipate risks in the moments preceding a potential incident rather than merely reacting after the fact. By embedding this foresight into the system, hazards are detected earlier than with traditional methods, leveraging real-time spatial data and predictive algorithms to provide actionable warnings that enable preemptive interventions – such as redirecting workers or halting machinery – to avert the near-miss before it fully manifests onsite.

Furthermore, by integrating human posture prediction, our near-miss detection system achieves a higher level of precision. Rather than relying solely on a worker's central trajectory, this approach triggers alerts whenever any body part encroaches on a hazardous zone or protective envelope. By incorporating joint-level posture estimation, the digital twin system tracks specific anatomical regions – such as limbs, head, or torso – in real time, enabling

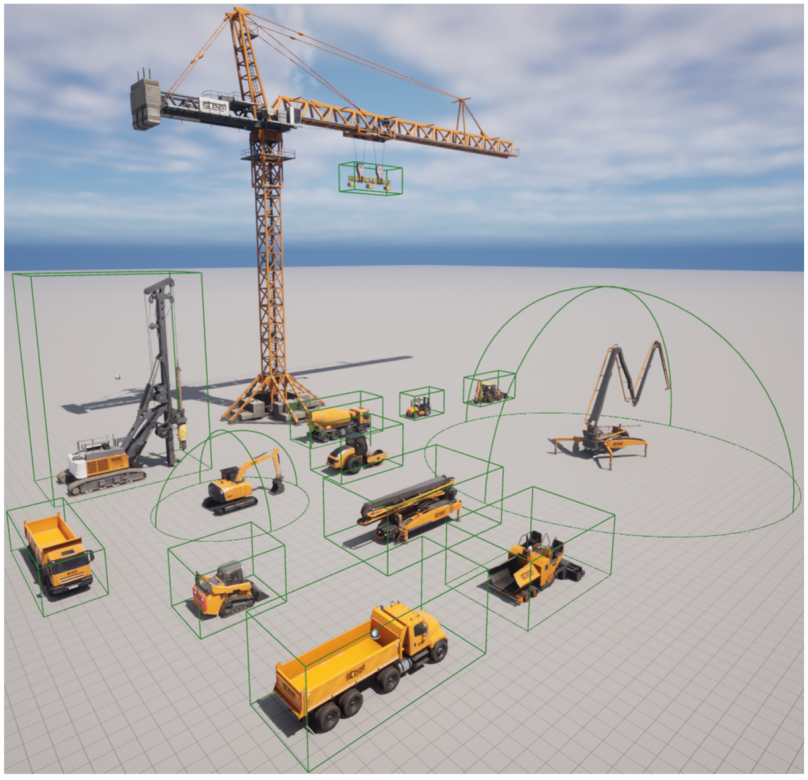

Figure 7. Protective boundary of dynamic elements. Source: Authors' own work

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detection of potential collisions at a granular level. This posture-aware monitoring captures incidents where even minor body parts approach danger zones, a sensitivity that surpasses traditional trajectory-based methods. Consequently, the system provides a more accurate and responsive tool for safety management, empowering site supervisors with detailed risk alerts and sufficient lead time to implement preventive measures, thereby reducing incident likelihood more effectively.

3.4 Digital twin visualization of near-misses

The digital twin system is fully deployed on the Unreal Engine (UE) platform, providing a dynamic, real-time visualization of the construction site's virtual twin alongside the live conditions at the physical site. Through this digital twin, both predictive and historical near-miss data are visualized, offering site managers and safety officers an immediate and comprehensive view of potential and actual incidents.

The system continuously monitors the positions and postures of workers and other dynamic elements within the virtual environment, forecasting potential near-misses based on real-time data. When a near-miss event is predicted, a visual alert is generated within the digital twin, immediately notifying on-site and remote staff of potential hazards. This real-time alerting system enables proactive intervention, reducing the likelihood of incidents by providing sufficient time for preventive actions.

Beyond immediate alerts, the digital twin logs and aggregates near-miss events, capturing details such as location, timestamp, and frequency. These records are visualized through interactive dashboards on the UE platform, enabling detailed safety analyses and long-term risk assessments. By mapping the frequency and spatial distribution of near-misses, the system reveals recurring hazards and operational patterns that may warrant intervention. This data-driven insight empowers managers to prioritize resources, refine safety protocols, and implement targeted measures, fostering a safer and more efficient construction environment.

4. Pilot study

Our proposed digital twin-based framework is designed to enhance proactive safety management on construction sites by focusing on predicting near-miss incidents through the forecasting of worker trajectories and postures. This innovative system integrates both static and dynamic data sources to construct an accurate digital representation of the construction environment. This comprehensive digital model allows for effective monitoring and prediction of potential hazards, thereby significantly improving the safety measures implemented on-site.

4.1 Implementation details

To enable real-time tracking of dynamic elements on the construction site, binocular cameras are strategically positioned at key vantage points across the area. These cameras continuously capture the spatial movements of workers, machinery, and other dynamic objects, generating a live stream of positional data critical for safety monitoring. The coordinates of each identified dynamic element are referenced to the origin of the respective binocular camera, establishing a localized frame of reference that anchors the virtual representation within the digital twin system.

Accurate reconstruction of these elements in the digital twin environment depends on precise knowledge of the cameras' spatial configuration. The current position and orientation of each binocular camera are meticulously documented. High-precision Real-Time Kinematic (RTK) GPS receivers and laser-based total stations determine the exact 3D position (x, y, z) of each camera relative to the site's coordinate system. Inertial Measurement Units (IMUs) integrated with the cameras capture their angular orientation (pitch, yaw, roll) with sub-degree

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accuracy. This transformation process converts the camera-centric coordinates into a unified global coordinate system aligned with the digital twin, ensuring seamless integration of real-time data into the virtual model.

The entire system is hosted on a computing platform equipped with an Intel® Core™ i7-14700K processor running at 3.40 GHz. This platform is complemented by a Nvidia 4090 GPU, which enhances the system's capability to handle complex calculations required for efficient processing. The operating system is Windows 10 Professional, and the system is fortified with 128GB of RAM, ensuring efficient data processing and management. This suitable setup facilitates seamless integration and functioning of the digital twin system, enabling accurate monitoring and analysis of construction activities.

4.2 Algorithm training

The training of our human motion prediction model utilized the widely recognized Human3.6M dataset (Ionescu *et al.*, 2014). This dataset comprises the 3D joint positions of seven actors engaged in 15 distinct activities, such as walking, sitting, and smoking. While it has traditionally served as a benchmark for human pose estimation, we adapt it for human trajectory prediction as well. For our trajectory predictions, we specifically extracted the hip trajectories of each actor, while for pose prediction, we focused on the relative positions of all other joints with respect to the stationary hip. In conventional practices, the frame rate of this dataset is often reduced from 50 Hz to 25 Hz; however, we opted for a frame rate of 10 Hz, which is more suitable for construction applications.

The model was developed using PyTorch as the primary deep learning framework, trained over 250 epochs with the AdamW optimizer, an initial learning rate of 10^{-4}, and a batch size of 16. A warm-up phase was implemented during the initial 10,000 steps, gradually increasing the learning rate from zero to 10^{-4}, which contributed to enhanced training stability.

To assess the effectiveness of the prediction algorithms, we utilize the following two commonly used metrics to measure the accuracy of motion prediction:

(1) Average Displacement Error(ADE):ADE calculates the average Euclidean distance between the actual values and the predicted values. It is computed by measuring the distance between the actual position and the corresponding predicted position at each step of the prediction, and then taking the average of these distances. The formula is typically expressed as:

$$ADE = \frac{1}{T}\sum_{t=1}^{T} \|p_t - \widehat{p}_t\| \tag{6}$$

where: T is the number of time steps for the prediction, p_t is the actual position at step t, $\widehat{p}_t$ is the predicted position at the same time step, $\|\cdot\|$ denotes the Euclidean distance. ADE reflects the average prediction accuracy of the model over the entire prediction process.

(2) Final Displacement Error(FDE):FDE refers to the Euclidean distance between the actual value of the target position and the predicted value at the final prediction step. It focuses solely on the accuracy of the last step. The calculation can be expressed using the following formula:

$$FDE = \|p_t - \widehat{p}_t\| \tag{7}$$

where: p_t is the actual position at the last time step, $\widehat{p}_t$ is the predicted position at the last step. FDE is primarily used to assess the accuracy of the model at the final step, which is particularly important in applications that require precise arrival at a specific location.

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On the test dataset, the prediction algorithm achieved an ADE of 0.134 and an FDE of 0.273 for trajectories. For human posture prediction, it achieved an ADE of 0.501 and an FDE of 0.752. These results demonstrate that the model is capable of generating temporally stable and spatially accurate forecasts of both body motion and limb configuration.

4.3 Application effects

To assess the effectiveness of the digital twin system in predicting near-miss incidents, we implemented it in active construction sites characterized by diverse environments and dynamic risk factors. The primary objective was to evaluate its practical applicability in identifying potential hazards through motion forecasting and near-miss prediction. The system was deployed across three distinct high-risk areas within the construction sites: the forklift path zone, the floor opening zone, and the material lifting zone. Each of these areas presents unique safety challenges, such as worker-vehicle interactions, fall hazards, and risks associated with overhead lifting activities.

To illustrate the working principle of the proposed Digital Twin-Enabled 3D Near-Miss Prediction Method more clearly, the workflow is exemplified by the floor opening zone, as shown in Figure 8. Field photographs captured by unmanned aerial vehicles (UAVs) are modeled into a BIM representation, which is regularly updated based on the latest inspection images. After being imported into Unreal Engine, the BIM model is used to establish an initial static digital twin scene.

A pair of binocular cameras installed on-site continuously capture left and right images. Through stereo matching, these images are processed to generate a depth map, thereby acquiring the spatial depth information of the scene. The left image is further utilized for object detection, object tracking, and human pose estimation. The detected human keypoints are then fed into a trained prediction model to forecast human posture and trajectory.

Subsequently, the obtained depth-enhanced object data, together with the pre-prediction human posture and location, as well as the post-prediction human pose and trajectory

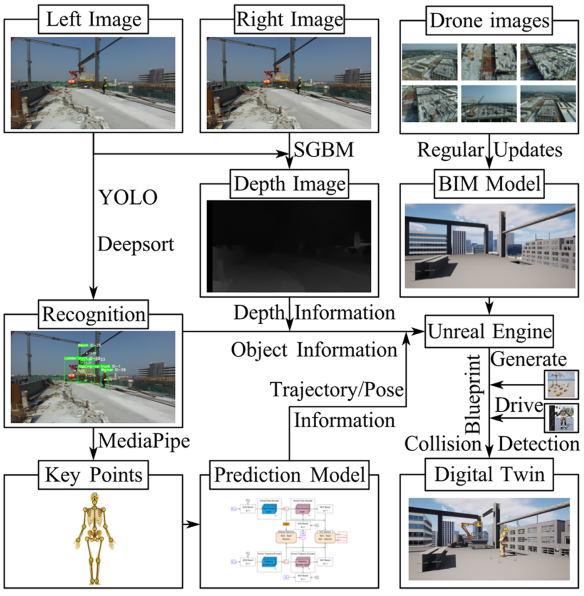

Figure 8. Workflow of the proposed method exemplified by the floor opening zone. Source: Authors' own work

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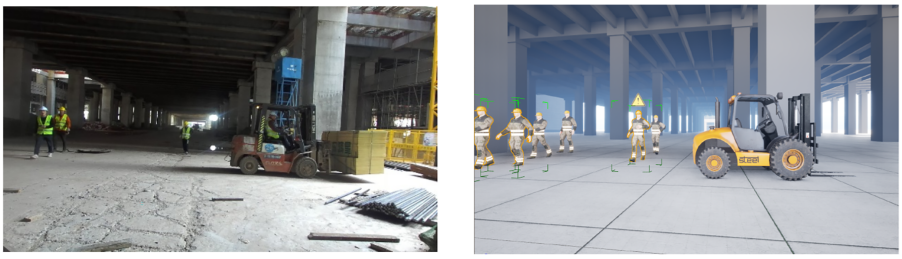
information, are transmitted to the digital twin platform for real-time scene generation, model updating, and human model animation. Finally, 3D collision detection is performed within the digital twin environment to enable near-miss prediction and early warning in three-dimensional space.

The interface of our Digital Twin-Enabled 3D Near-Miss Prediction Method is illustrated in Figure 9. In each zone, the digital twin system uses binocular camera data to identify worker movements, displaying each predicted worker trajectory and posture within a green cuboid frame in the digital model. When a potential near-miss is detected, a hazard warning signal is displayed above the worker in the interface, alerting site managers to the elevated risk.

For example, in Figure 10, subfigure (a) shows the actual scene in the forklift pathway zone, while subfigure (b) displays the corresponding digital twin view of the construction site. In the digital twin, workers predicted to potentially intersect with the forklift's path are highlighted within green cuboid frames. A hazard warning signal appears above the worker's virtual representation, and the system issues a preemptive alert to prompt intervention.

Figure 9. Interface of our digital twin-enabled 3D near-miss prediction method. Source: Authors' own work



Physical world
(a)

Digital twin world
(b)

Figure 10. The forklift path zone. Source: Authors' own work

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Similarly, in Figure 11, subfigure (a) depicts the real-world scene in the floor opening zone, and subfigure (b) illustrates the digital twin environment. Here, the system predicts a worker moving towards an unattended floor edge, posing a fall risk. The system provides an early warning to prevent potential falls by alerting site personnel.

In Figure 12, subfigure (a) shows the actual material lifting zone, while subfigure (b) represents the digital twin scene. In this case, the digital twin forecasts that the worker will remain stationary, with the predicted and actual worker positions overlapping, indicating no immediate near-miss risk. However, the system continuously monitors the worker's proximity to suspended loads, issuing a warning if they approach a hazardous area.

In addition, to establish a reliable ground truth for evaluation, experienced on-site safety managers were invited to assess and annotate near-miss events throughout the experiment. Their expert judgments were recorded and later used as reference standards to assess the system's predictive accuracy and reliability in real-world conditions.

4.4 Evaluation of results

Based on the results from field tests, a comprehensive evaluation of the digital twin system's performance was conducted to assess the accuracy of near-misses prediction. The evaluation primarily relied on the precision and recall rates of target detection, as detailed in Formulas (8) and (9). Precision measures the proportion of correctly predicted near-misses relative to all predicted near-misses, providing an indicator of the system's accuracy in hazard identification.

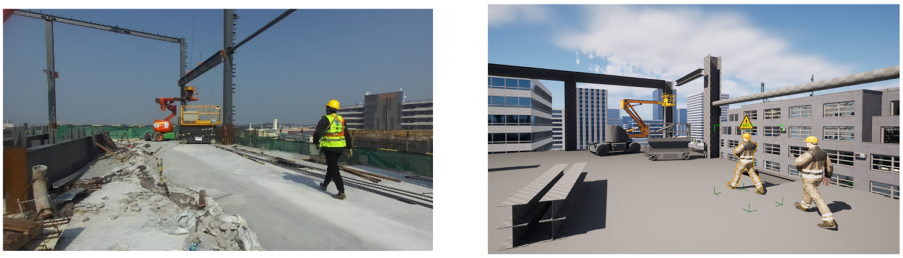

Figure 11. The floor opening zone. Source: Authors' own work

Figure 12. The material lifting zone. Source: Authors' own work

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Recall evaluates the proportion of actual near-misses that the system successfully identified, reflecting its sensitivity in detecting potential risks. These metrics provide insight into the system's ability to accurately identify and predict near-misses on construction sites.

$$Precision = \frac{N_{correct}}{N_{correct} + N_{incorrect}} \tag{8}$$

$$Recall = \frac{N_{correct}}{N_{correct} + N_{undetected}} \tag{9}$$

where $N_{correct}$ represents the number of near-misses accurately predicted by the digital twin, $N_{incorrect}$ represents the number of near-misses inaccurately predicted, and $N_{undetected}$ indicates the number of actual near-misses that were not predicted by the digital twin system.

As shown in Table 1, the digital twin system achieved overall precision and recall rates above 90% in predicting near-misses across the three designated zones, demonstrating its effectiveness in accurately identifying potential hazards. In the forklift path zone, the system's precision and recall were slightly lower compared to the other two areas. This may be attributed to the high worker density and frequent movements in this zone, which can result in occlusions among workers, making it challenging for the binocular cameras to capture all individuals accurately. In contrast, the digital twin system performed best in the floor opening zone. This enhanced performance is likely due to the optimal placement of the binocular cameras, which provides unobstructed views of workers in this area. Additionally, better lighting conditions in the floor opening zone further supported the system's computer vision-based capabilities.

While the digital twin system demonstrated high accuracy across three high-risk construction zones, several limitations were observed. First, the system's performance is influenced by visual occlusions, lighting variability, and the positioning of binocular cameras, which may affect the reliability of real-time detection and tracking. Second, the prediction model was trained on the Human3.6M dataset, which may not fully capture the variability of real-world construction postures, leading to a potential generalization gap.

Although the validation scenarios were relatively simple, the results affirm the feasibility of using the digital twin system to predict near-misses by forecasting workers' trajectories and postures. These findings provide a strong foundation for further development of proactive safety management tools in complex construction environments. In future work, we aim to extend the system's capabilities by exploring task-specific risk factors – such as differentiated hazard thresholds for forklift operation zones versus floor opening areas – and incorporating adaptive mechanisms based on dynamic worker–machine interaction patterns. Such enhancements are expected to improve the system's contextual sensitivity and operational robustness in diverse and evolving construction scenarios.

Table 1. Results of near-misses prediction

Performance metric	The forklift path zone	The floor opening zone	The material lifting zone	Total
Correctly predicted	189	273	544	1006
Mis-predicted	32	1	39	72
Not predicted	26	0	15	41
Precision	85.52%	99.64%	93.31%	93.32%
Recall	87.91%	100.00%	97.32%	96.08%

Source(s): Authors' own work

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5. Conclusions and future work

In this study, we successfully developed and implemented a proactive safety management system utilizing digital twin technology to enhance construction site safety by predicting near-miss incidents. By integrating multi-source data – including information about buildings, construction personnel, materials, and machinery – we established a digital twin model capable of accurately mirroring the real status of construction sites. This model enables precise prediction of construction workers' trajectories and postures, facilitating early warnings of potential near-miss events. The system's effectiveness was validated through field tests conducted in three high-risk areas: the forklift path zone, the floor opening zone, and the material lifting zone. In these applications, the system demonstrated high prediction accuracy, highlighting its potential to improve safety management on construction sites.

The innovation of this study is reflected in three key aspects:

(1) From 2D to 3D: Our approach transcends the limitations of traditional 2D safety management systems by extending analysis and monitoring into three-dimensional space, enabling a more precise representation of worker movements and potential hazards.

(2) Transition from Detection to In-Process Prediction: Unlike conventional methods that either focus on post-event detection or rely on historical data to infer future probabilities, our method introduces a proactive, in-process prediction mechanism. It dynamically models the real-time trajectories and postures of workers as construction activities unfold, forecasting potential near-miss events during the process rather than merely assessing historical trends. This allows for early identification of evolving hazards and timely interventions to prevent accidents.

(3) 3D Visualization of Near-Misses: Through the integration of digital twin technology, we realized 3D visualization of predicted near-misses, providing intuitive, actionable insights that empower safety managers to respond effectively and enhance overall safety management practices.

Despite our achievements, this study has some limitations. Firstly, the limited amount of data affects the generalization ability of the model. Although we used the widely recognized Human3.6M dataset for training, for the complex and variable construction scenarios, more on-site data may be required to further optimize the model. Secondly, as the proposed method in our study is based on a computer vision solution, it may be affected by factors such as occlusions in the scene and poor lighting conditions.

In addition to data and perception-related limitations, we also recognize that scalability poses a significant challenge for the widespread application of the proposed system. As construction sites expand in size and complexity, deploying and maintaining a dense network of binocular cameras and UAVs for real-time data collection becomes increasingly costly and technically demanding. Larger-scale environments may also generate high volumes of spatiotemporal data, which could overwhelm existing computing resources, delay prediction responses, and compromise system performance under limited bandwidth or unstable network conditions. Furthermore, prediction models trained on small-scale or controlled environments may require site-specific adaptation when applied to different spatial layouts, worker behaviors, or operational protocols.

In response to the above limitations, future research can be carried out in the following directions. First, integrate more data sources, such as introducing more types of sensor data (e.g. environmental sensors, physiological sensors, etc.) to obtain more comprehensive construction site information and further improve the accuracy of near-miss incident prediction. Second, further optimize the model by exploring more advanced algorithms to handle human pose and trajectory prediction in complex environments and improve the robustness of the model against interference factors such as occlusion and illumination

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changes. Third, to address scalability concerns, investigate lightweight and modular system architectures that support edge computing and distributed deployment, as well as domain-adaptive learning techniques to enable model generalization across heterogeneous construction sites. Finally, conduct more extensive validation in different types and scales of construction sites to ensure the applicability and effectiveness of the system, thereby providing more universal solutions for safety management in the construction industry. Through these future research directions, we expect to further enhance the application value of digital twin technology in construction safety management and promote the development of the construction industry towards a safer and more efficient direction.

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